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Ray tracing (Descartes & Newton)

In 1637, Descartes used ray tracing to understand the formation of primary and secondary rainbows. Newton subsequently extended the theory to explain the colours of rainbow. Although ray tracing cannot match the accuracy of Mie theory, it remains valuable because it can provide intuitive explanations of many features of rainbows - without demanding the use of complicated mathematics.

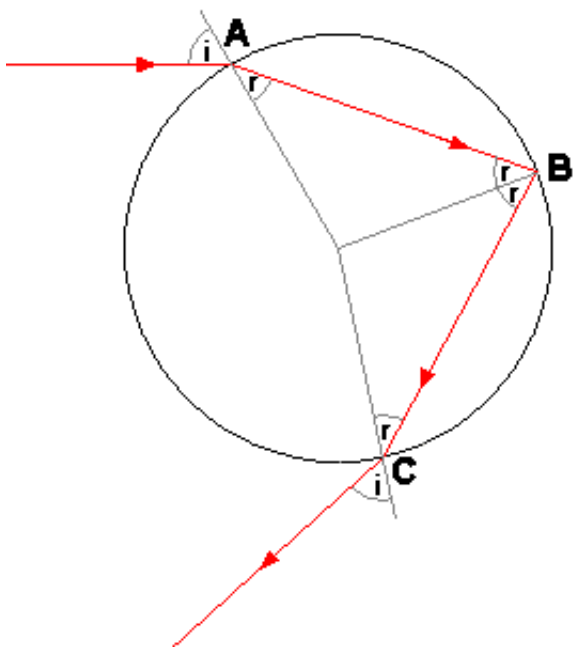

Fig. 1 Basic geometry of the primary rainbow

The primary rainbow is the result of light rays which have suffered one internal reflection within the sphere ($p = 2$) - as shown in Fig. 1 where the ray of light impacts a spherical drop of water at point A with angle of incidence i. After entering the drop with angle of refraction r, it is reflected at point B and leaves the drop at point C at an angle relative to the original direction of $180° + 2\,i - 4\,r$. This scattering angle θ can be generalised for other values of p as shown below:

for $p = 0$, $\theta = 2\,i - 180°$

for $p = 1$, $\theta = 2\,i - 2\,r$

for $p > 1$, $\theta = 2\,i - 2\,p\,r + (p - 1)\,180°$

If the refractive index of the sphere is n, Snell's Law defines the relationship between angle r and angle i as:

$n = \sin[i] / \sin[r]$.

As r is dependent on i, the scattering angle θ can be generalised for other values of p as shown below:

for $p = 0$, $\theta = 2\,i - 180°$

for $p = 1$, $\theta = 2\,i - 2 \arcsin[\sin[i] / n]$

for $p > 1$, $\theta = 2\,i - 2\,p \arcsin[\sin[i] / n] + (p - 1)\,180°$

H. C. van de Hulst's book "*Light scattering by small particles*" contains an excellent diagram (Fig. 41 on page 229) showing how the scattering angle varies with the angle of incidence i for various values of p. Fig. 2 below shows an equivalent diagram for refractive index $n = 1.33257$.

Due to symmetry, the scattering angle θ is normally defined in the range 0° to 180°, Fig. 3 compresses the results for $p < 4$ into the range 0 to 180°. If you want to convert any angle α into the range 0° to 180°, simply use the functions $\arccos[\cos[\alpha]]$ on your calculator.

Fig. 3 indicates that, for $n = 1.33257$, a minimum occurs for $p = 2$ at a scattering angle θ of 137.9° (the primary rainbow angle), whilst a maximum occurs for $p = 3$ at $\theta = 127.9°$ (the secondary rainbow angle). The very slow variation of scattering angle near these minima or maxima has the effect of concentrating the scattered light, thus generating intense narrow beams of light close to the rainbow angle.

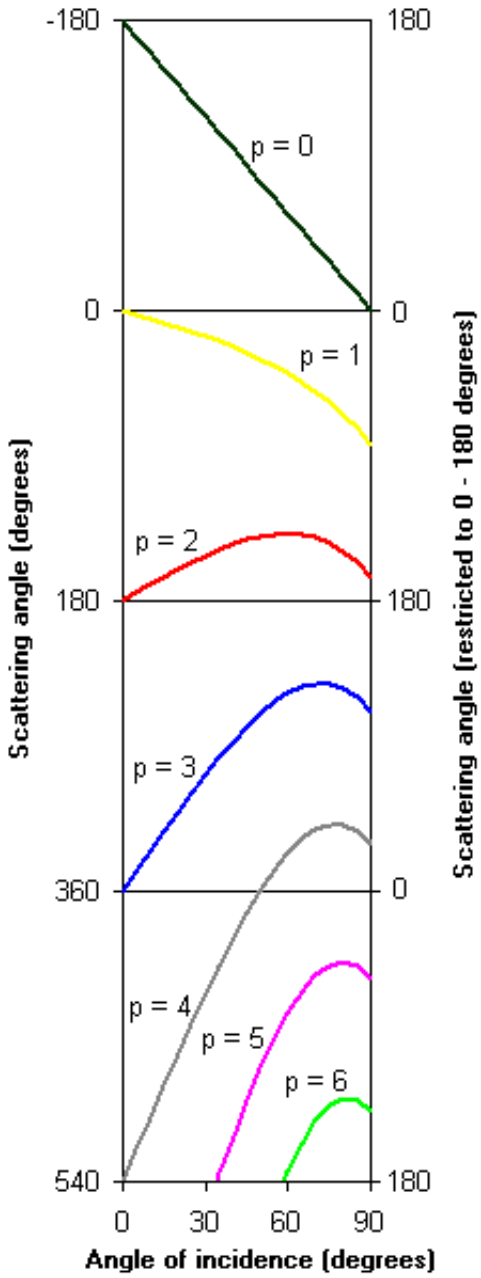

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Fig. 2 Scattering angle θ as a function of angle of incidence *i* for specified values of *p*

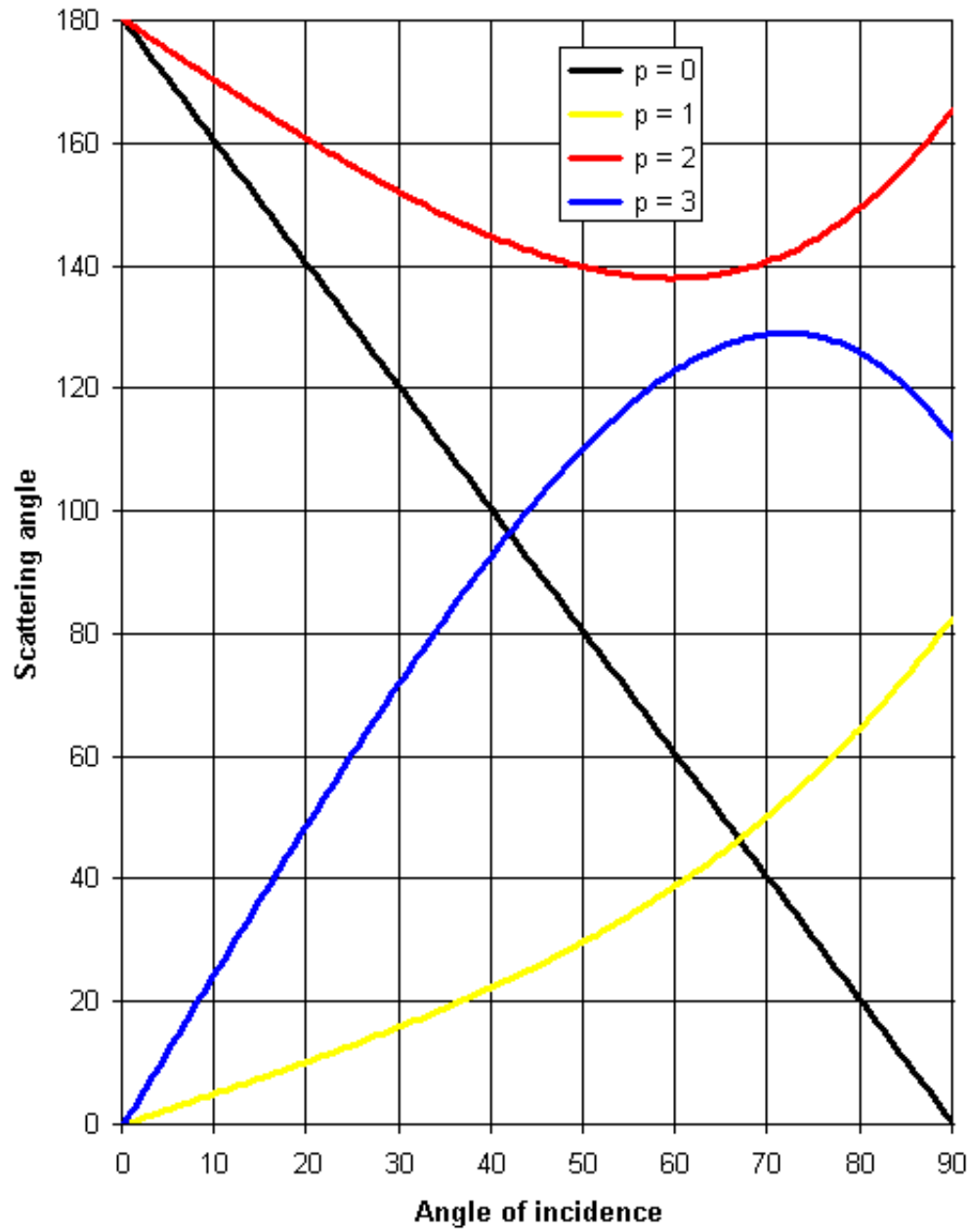

Fig. 3 Scattering angle θ as a function of angle of incidence *i* for $p < 4$

The angles of incidence *i* corresponding to the geometric rainbow angles are defined by:

$$i = \arccos\left[((n^2 - 1) / (p^2 - 1))^{0.5}\right]$$

For example, for $n = 1.33257$ and $p = 2$, this equation gives $i = 59.44°$. Using the other equations above, you will find that $r = 40.25°$ and that the resulting rainbow angle is at $\theta = 137.86°$. The following table shows the geometric rainbow angles for values of *p* between 2 and 7 when $n = 1.33257$.

n = 1.33257	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$	$p = 7$
Angle of incidence i	59.44°	71.86°	76.86°	79.64°	81.44°	82.70°
Angle of refraction r	40.25°	45.49°	46.95°	47.58°	47.91°	48.10°
Impact parameter $b = \sin[i]$	0.861057	0.950280	0.973799	0.983706	0.988856	0.991886
Rainbow angle	137.86°	129.22°	41.90°	43.50°	127.98°	148.04°

When the impact parameter $b = 0$, the incident ray is a "central" ray passing through the centre of the sphere. When $b = 1$, the incident ray is an "edge" ray which is tangential to the sphere. Even in the case of $p = 2$ rainbow, as shown in Fig. 4 below, the impact parameter of 0.861057 corresponds to a ray which is incident close to the edge the sphere. The above table shows that, as p is increased, the impact parameters associated with each rainbow progressively approach 1 (i.e. all rainbows are caused by incident rays which arrive close to the edge of the sphere).

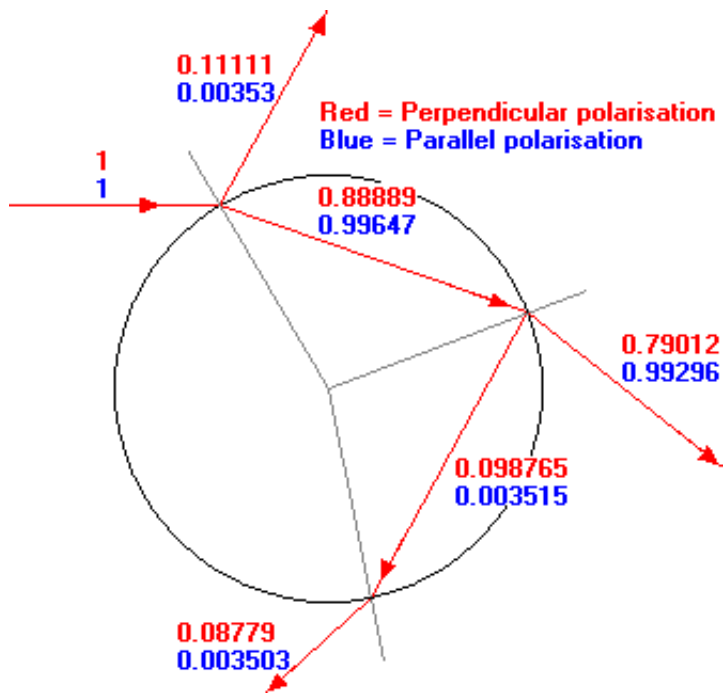

Fig. 4 Relative intensities of of rays forming the primary rainbow ($p = 2$) for $n = 1.33257$

The relative intensities of the various rays can be calculated by using the Fresnel reflection coefficients. The rays shown in Fig. 4 above correspond to the rainbow angle for $p = 2$ rays (i.e. the primary rainbow). Although the incident light has an intensity of 1 for both polarisations, Fig. 4 indicates that the intensity of

the primary rainbow is significantly greater for perpendicular polarisation than for parallel polarisation.

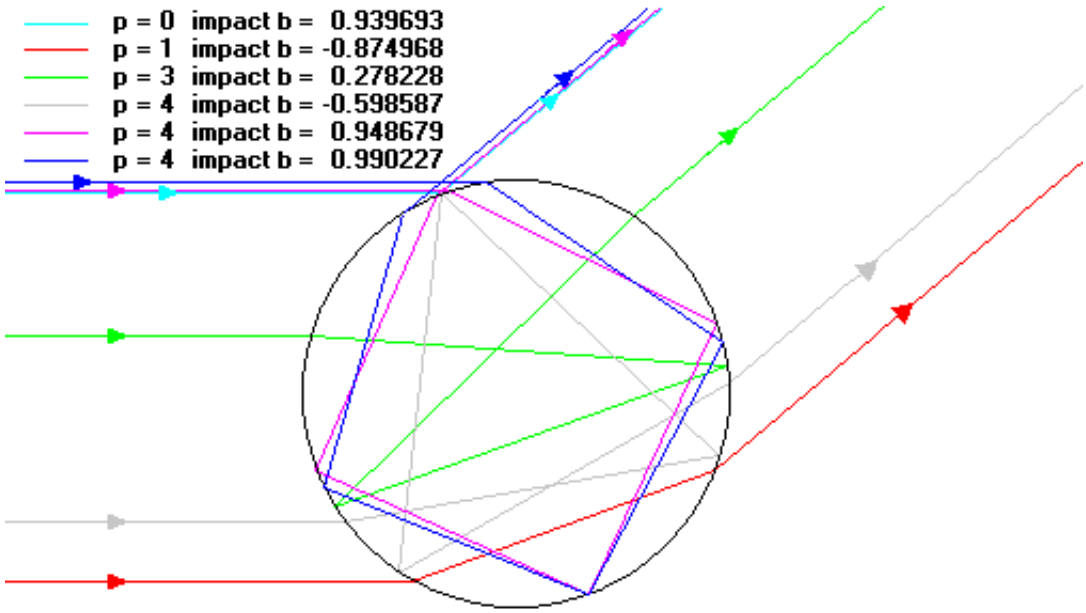

Fig. 5 Ray tracing showing various parallel rays arriving from the left and leaving the sphere at a scattering angle θ of 40° (for refractive index of n = 1.33257)

A beam of light arriving from the left of Fig. 5 can be modeled as a bundle of parallel rays which are reflected and/or refracted by the sphere. Instead of showing all of the rays, Fig. 5 shows only a few selected rays - all of which emerge at a scattering angle θ of 40°. As the incident light is coherent, the intensity of the light scattered in a particular direction (e.g. θ = 40°) is dependent on the relative amplitudes of each of the rays shown in Fig. 5 and on the phase delay caused by various propagation paths through the sphere.

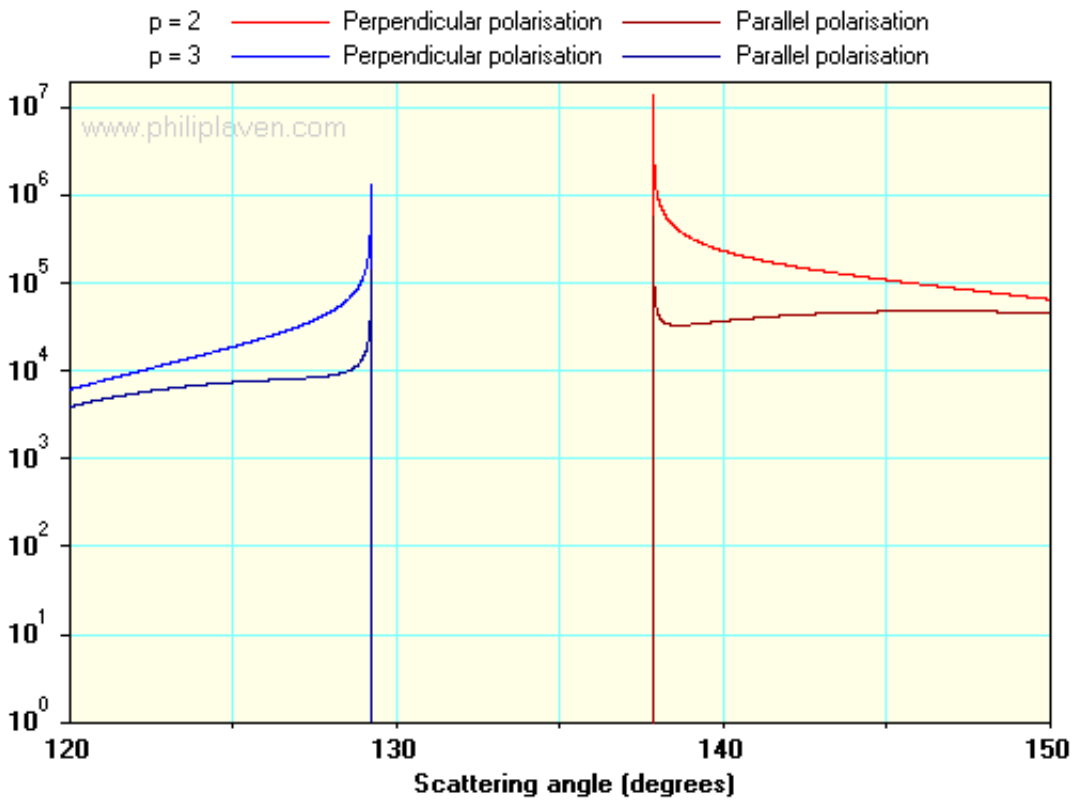

Fig. 6 Results of ray tracing showing the intensity of the primary rainbow ($p = 2$) and the secondary rainbow ($p = 3$) for $n = 1.33257$ and radius $R = 100$ μm

Note that ray tracing predicts infinite intensity at each of the rainbow angles (e.g.for $p = 2$, $p = 3$, $p = 4$, etc.). Furthermore, Fig. 6 shows that ray tracing predicts zero intensity between the two rainbow angles of 127.9° and 137.9° because no geometric ray for $p = 2$ or $p = 3$ can be scattered into this zone - corresponding to Alexander's dark band. Note that the shape of the scattering pattern predicted by ray tracing is independent of the radius R of the scattering sphere, but the intensity is proportional to R^2.

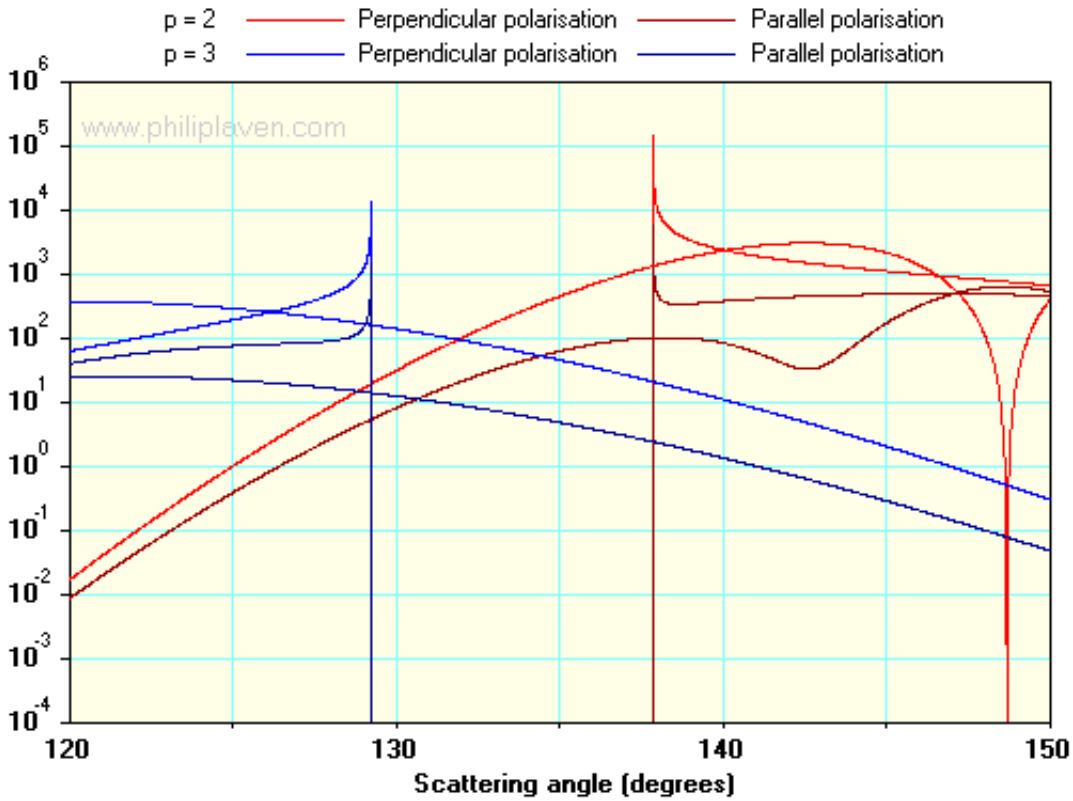

Fig. 7 Comparison of ray tracing and Airy theory for the primary rainbow (p = 2) and the secondary rainbow (p = 3) for n = 1.33257 and **radius R = 10 μm**

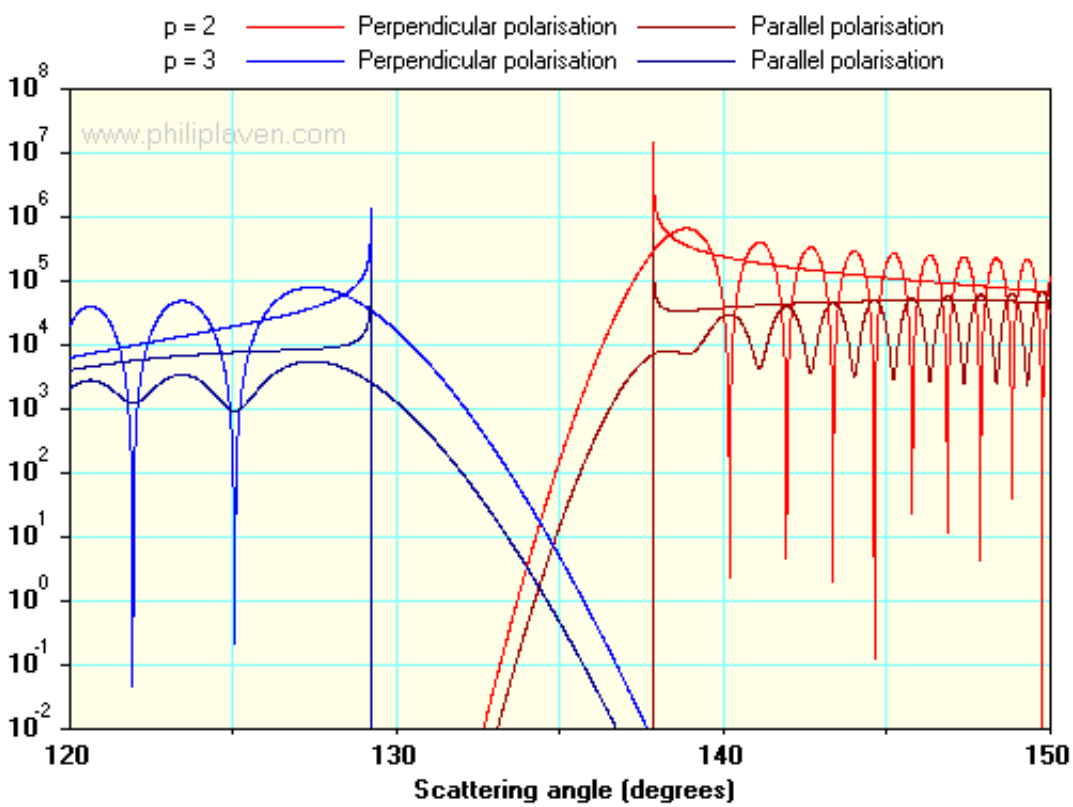

Fig. 8 Comparison of ray tracing and Airy theory for the primary rainbow (p = 2) and the secondary rainbow (p = 3) for v = 1.33257 and **radius *R* = 100 *μ*m**

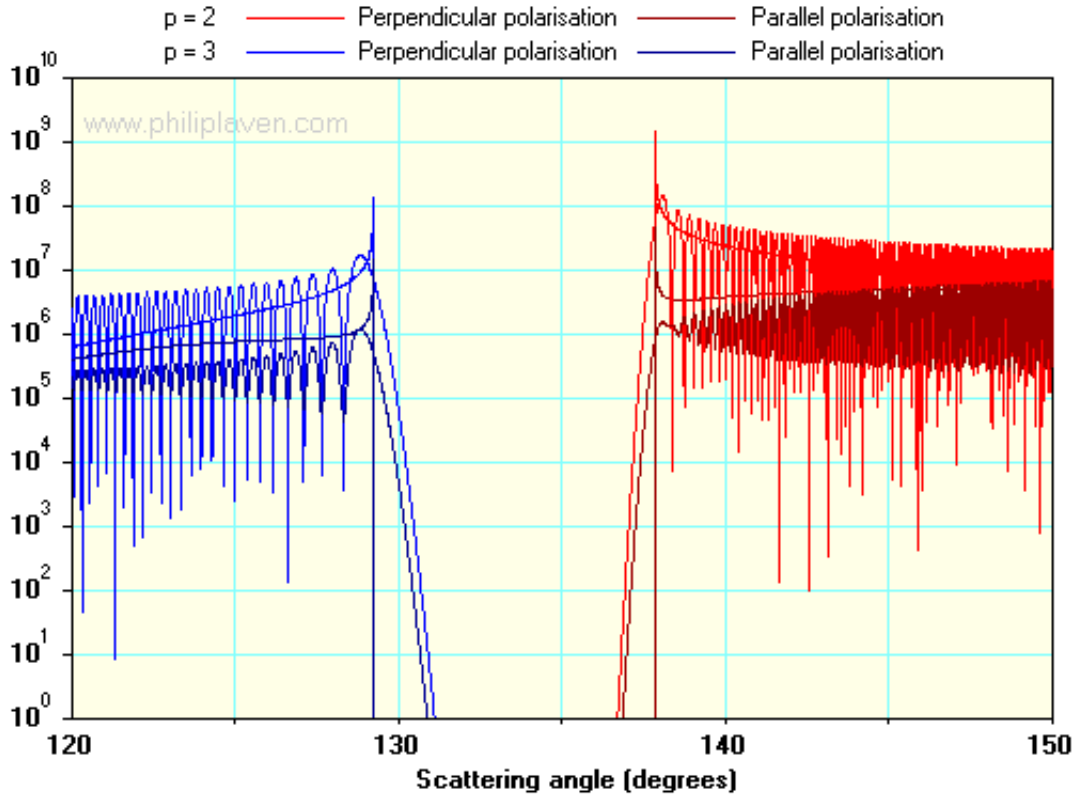

Fig. 9 Comparison of ray tracing and Airy theory for the primary rainbow (p = 2) and the secondary rainbow (p = 3) for n = 1.33257 and **radius *R* = 1000 *μ*m**

By contrast, Airy theory predicts very different scattering patterns for different values of radius - as shown in Figs. 7 - 9 for radius R = 10 μm, 100 μm and 1000 μm respectively. Note that Airy's maximum intensity does not coincide with the geometric rainbow angle predicted by ray tracing: the error becomes small only when the radius of the sphere is very large (e.g. $R > 1000$ μm) Airy theory also shows significant intensity in Alexander's dark band between the primary and secondary rainbow. Finally, the supernumerary arcs predicted by Airy theory are not reproduced by ray tracing. Because of these obvious shortcomings, ray tracing has generally been dismissed as a useful mathematical model. However, as will be shown on the next page, some refinements of ray tracing can yield some interesting results

MiePlot offers the option of calculations based on ray tracing, using the methods developed by Descartes/Newton and Young.

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